Gutters & Downpipes

User guide · IngeTrazo plugin · `gutters_downpipes.py` · Ctrl+Alt+G

Builds full-size, hollow or solid gutters, downpipes and fittings, each in its own group, and joins every new part onto the open end of the last. Its window stays open while you work, so you can add a whole run part by part.

Installing and updating

1. Open File Explorer, paste `C:\Users\eddy_\AppData\Roaming\ingetrazo\plugins` into the address bar and press Enter.
2. Copy `gutters_downpipes.py` into that folder. When updating, choose **Replace the file**.
3. Restart IngeTrazo. Settings are remembered only while IngeTrazo is open, so a restart also clears old settings.

Keep only one copy of the plugin in the folder. Two copies under different names may both load.

Shortcut: Ctrl+Alt+G opens the window, or brings it to the front if it is already open.

From the Python console: paste the file's code in, select what the plugin needs, then type `run()`. If it can't find the 3D view by itself, use `run(viewport)`.

The window

Area	What it does
Top line	Says where the next part will go, for example "joins onto the far end of: Gutter length"
Solid (for 3D printing)	Fills gutters and pipes in, with no hollow inside
Preset	Picks a saved set of sizes; **Save as**, **Delete** and **Open folder** manage your own presets
Part list	Gutter parts and downpipe parts
Gutter / Downpipe	Profile or shape, width or diameter, depth and wall thickness, in mm
This part	A live 3D preview, plus the settings the chosen part uses
Start a gutter on a wall	Places the first gutter along the top of a wall you select
Brackets and clips	Automatic fascia brackets and pipe clips, and chunky fittings for printing
Buttons	**Add part**, **Use selected end**, **Start at the origin**, **Close**

In the preview, the brown panel is the wall and the pale grey part is the one the chosen fitting sits over.

All sizes are typed in millimetres. IngeTrazo stores geometry in metres whatever your display units, and the plugin converts for you.

Parts

Part	Settings it uses	Notes
Gutter length	Length	Gets fascia brackets automatically
Running outlet	Outlet length	Its spigot underneath takes a downpipe
Stop end	—	Closes a gutter run
Union bracket	Distance back from the end	Sits over the joint between two lengths
Fascia bracket	Distance back from the end	Its back plate sits on the wall
90° and 135° angles	Leg length	External corners turn towards the wall, internal corners away from it
Downpipe length	Length	Gets pipe clips automatically
Bend / offset bend	Turn angle, direction	Turn 67.5 = 112.5° offset bend, 87.5 = 92.5° bend, 45 = 135° bend
Branch (Y junction)	Turn angle, direction, branch length	The arm angles back up towards the incoming flow
Shoe	Direction	Kicks away from the wall by default; closes a downpipe run
Pipe socket / coupler	Distance back from the end	Sits over a pipe joint
Pipe clip	Distance back from the end	Its lug points at the wall

Gutter profiles are half-round, square-line, ogee and deep-flow. Downpipes are round or square.

Building a run

1. Choose a starting point. With nothing chosen, the first part goes at the origin; **New run - wall is** sets which side the wall is on. To start on a building, see **Starting on a wall** below.
2. Pick a part, check its sizes in the preview, and click **Add part**.
3. The next part joins onto the far end of the one you just added. Keep adding parts.
4. After a running outlet, choosing a downpipe part puts it on the outlet's spigot automatically.
5. A stop end or shoe finishes a run. The window then moves to the newest end still open, for example the outlet spigot, or the start of the gutter for the second stop end.

To join onto any other end, click that open end face in the model, then **Use selected end**. Unions, sockets and clips sit over a joint, and the next part still carries on from that joint.

You can move or rotate placed parts with IngeTrazo's own tools and still join onto them. If you undo a part, click an end and **Use selected end**, because the window still points at the removed part.

Starting on a wall

1. Select a wall face and click **Use selected wall**.
2. Pick a gutter part and click **Add part**. It runs along the top of the wall, starting from the left end as you look at the wall from outside.

Setting	Default	Effect
Below top of wall	25 mm	Gap from the top of the wall down to the top of the gutter
Extra gap from wall	0 mm	0 puts the brackets flat on the wall
Past left end / Past right end	0 mm	Positive runs past that end. Negative stops short of it, leaving room for an outlet, angle or stop end
Gutter length = wall length	On	The first gutter length fits the wall, including the two end settings
Wrong side?	Off	Tick it if the gutter appears on the inside of the wall

Brackets and clips

With **Add them evenly** ticked, each gutter length gets fascia brackets and each downpipe length gets pipe clips. One goes about 150 mm from each end, and the rest are spaced evenly between, no further apart than you set: 1000 mm for gutters and 1800 mm for pipes by default. A 5 m gutter gets 6 brackets; a 3 m downpipe gets 3 clips. They go in their own group, and a single undo removes them together with their length.

Chunky fittings (for 3D printing) makes every bracket, clip, union, socket and stop end thicker, 3× by default. It also closes the gap between each fitting and the part it fits over, and turns each fascia bracket into one solid block down to its wall plate. On a wall, the gutter moves out just enough for the chunkier brackets to stay flat against it.

For small-scale printing, tick both **Solid** and **Chunky fittings**, then scale a copy with 3D Print Scaler.

Presets

Four presets are built in:

- UK half-round 112 + round 68
- UK square-line 114 + square 65
- Deep-flow 117 + round 68
- Cast iron ogee 115 + round 75

Save as stores your current sizes and options as one `.json` file in Documents › IngeTrazo Presets › Gutters and Downpipes. **Open folder** opens that folder, so you can copy or share preset files.

Troubleshooting

Problem	Cause	Fix
A part comes out about 1000 times too big	An old version treated sizes as metres	Use the latest file and restart IngeTrazo
A new part doesn't join where expected	The window still points at an undone or deleted part	Click the right open end and use **Use selected end**
The gutter appears inside the wall	The wall face points inwards	Tick **Wrong side?** in Start a gutter on a wall

Problem	Cause	Fix
"Could not mark the open ends" in the console	IngeTrazo didn't keep the joining marker on the faces	That part can't be joined onto; add the next part from the origin and move it into place

Limits

- The running outlet has no hole cut in the gutter floor; its spigot sits underneath.
- Joints are butt joints, not overlapping sockets.
- Joining works only on ends made by this plugin.